

PES1UG24CS386
Roopa Sreedhar A
G

Architecture Justification
Clinical Trial Subject Recruitment Engine

We chose Layered Architecture (Presentation, Business, and Data layers) for the Clinical Trial Subject Recruitment Engine.

Two Scenario-Related Reasons

1. Regulatory traceability: The system must comply with 21 CFR Part 11 (NFR-001), which requires clear, auditable ownership of every action on a regulated record. A layered design isolates candidate-facing eligibility screening (Presentation), the eligibility, consent, and visit-compliance logic (Business), and record storage (Data) into distinct, independently verifiable responsibilities, making it far easier to demonstrate to auditors exactly where and how each decision and record change is produced.

2. A single, consistent source of truth: Clinical Researchers must be able to trust the eligibility result, consent status, and visit-compliance data they review (FR-003, FR-005). Routing every screening, consent, and compliance service through one shared Data Layer keeps subject records consistent, and avoids the data-synchronization and eventual-consistency risks that a distributed microservices split would introduce for a system of this size, where correctness of a single subject record matters more than independent scaling of services.

Security Advantage

All persistence and tamper-evident audit logging sit behind the Data Layer and are reached only through the Business Layer's dedicated Audit & Security Service. This creates one enforced choke point for authentication and audit-trail writes rather than several independently reachable data stores, so unauthorized or untraceable modification of a regulated record (NFR-001) has a single, well-guarded path to close off and monitor.

Performance Benefit

Because the Presentation Layer calls Business Layer services in-process rather than over a network, and all services share one consistent database rather than coordinating across independent services, eligibility screening and consent/compliance status lookups (FR-001, FR-002, FR-004) return with low, predictable latency. This directly supports the reliable, easy-to-use interface required by NFR-002, without the added network latency and coordination overhead a microservices approach would introduce.

Component Diagram

Clinical Trial Subject Recruitment Engine — UML Component Diagram (Layered Architecture)

